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FLIGHT DELAY PREDICTIONS

MSIS 502 Presentation



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INTRO AND PURPOSE

Examining 26 airlines (narrowed down to 9), with flight data collected since 2003, and flights departing from Sea-Tac Airport.

Visualize and analyze the number of issues attributed to the airline carrier (issues within the airline's control, such as maintenance problems, crew issues, or other operational reasons).

Apply business insights from the analysis and present conclusions

DATA CLEANING

- Saw carrier names wasn't consistent throughout the spreadsheet.
- Renamed the less common carrier names to the more common ones.
- Created a column called month year, since the data gave year in one column and month in another.
- Calculated percentage columns so we have a column to make predictions using linear regression, since the data was given as an aggregate.

carrier	carrier_name	
AA	American Airlines Inc.	175
	American Airlines Network	88
AS	Alaska Airlines Inc.	175
	Alaska Airlines Network	88
B6	JetBlue Airways	263
CO	Continental Air Lines Inc.	103

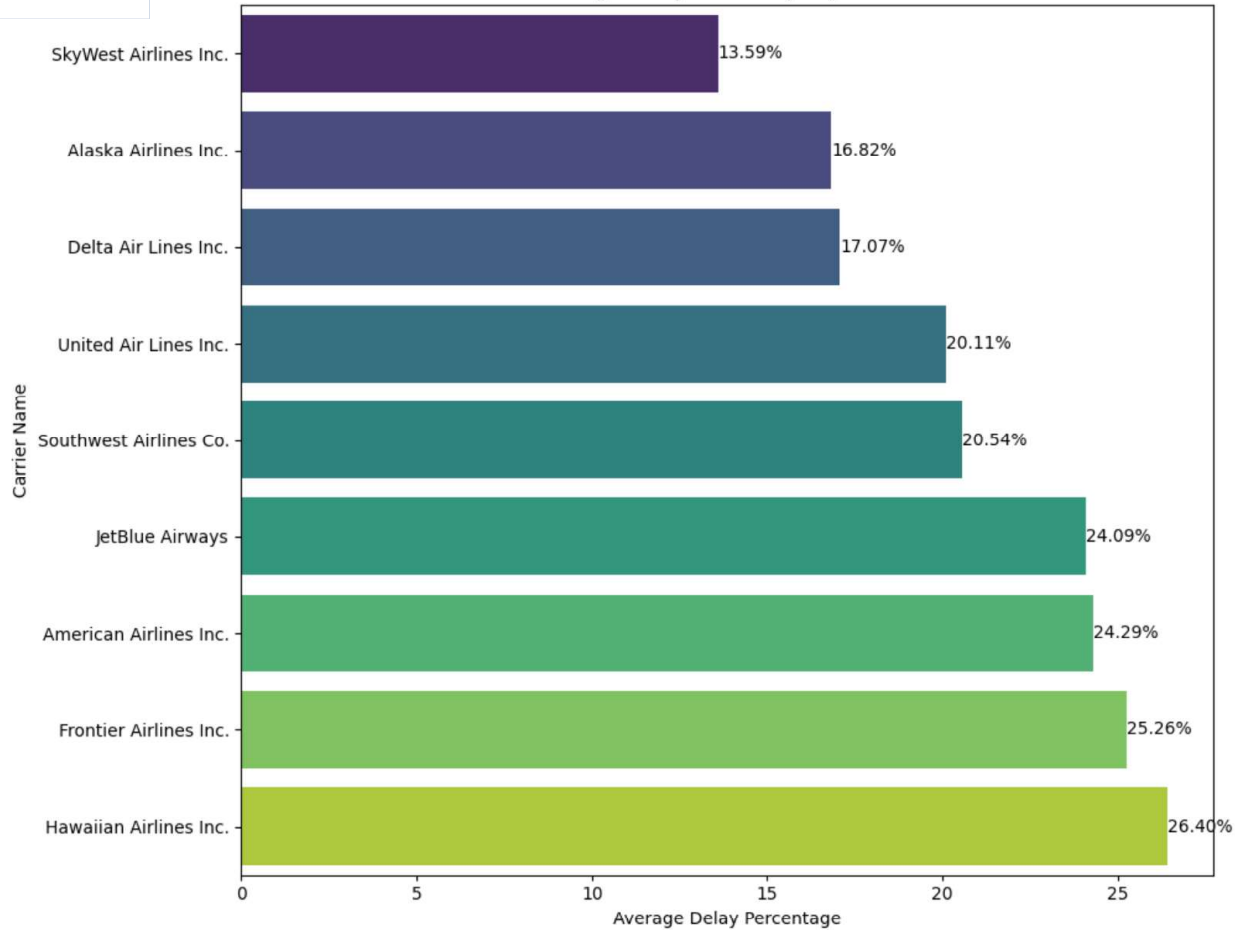
month_year	delay_percentage	cancel_percentage	diverted_percentage
2025-04-01	0.4037	0.0000	0.0
2025-04-01	0.1452	0.0161	0.0
2025-04-01	0.1101	0.0000	0.0
2025-04-01	0.1478	0.0041	0.0
2025-04-01	0.2111	0.0133	0.0



ANALYSIS AND INFERENCES

- Filter and plot the most relevant data that determine service reliability
- Focus visualizations on the top 9 major airlines for interpretability.
- Average delay by Carrier (%)
- Average cancel by Carrier (%)
- Average diversions by Carrier (%)

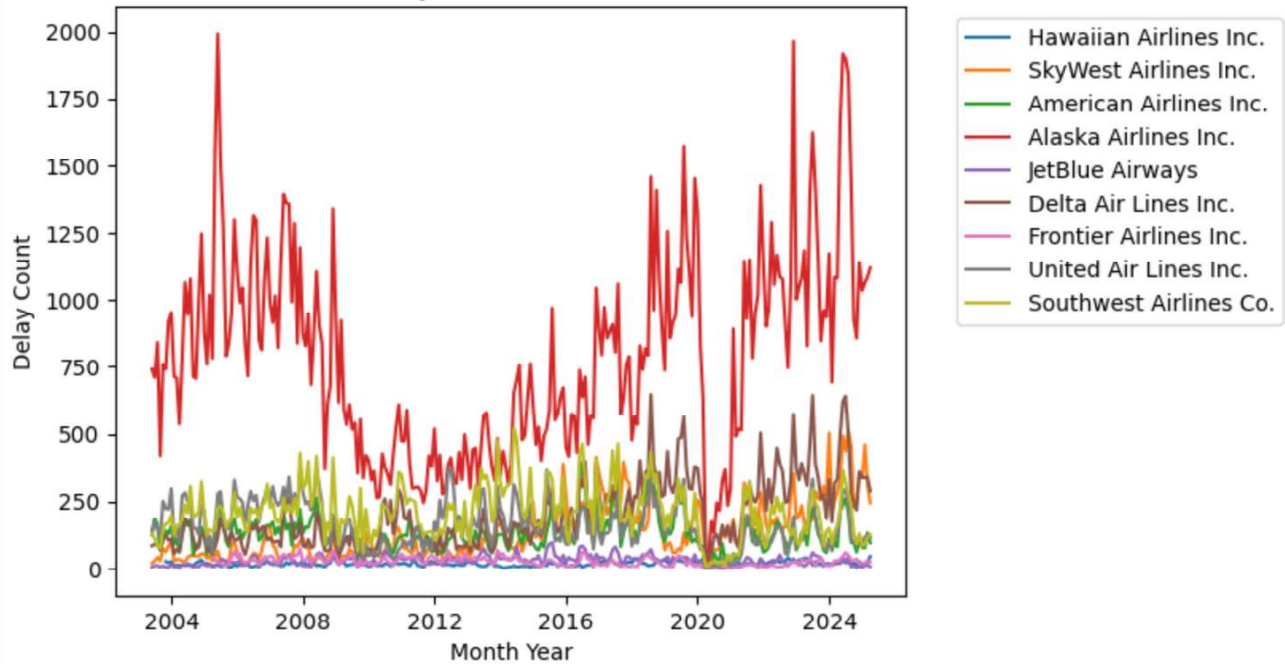
Average Delay Percentage by Carrier



DELAY ANALYSIS

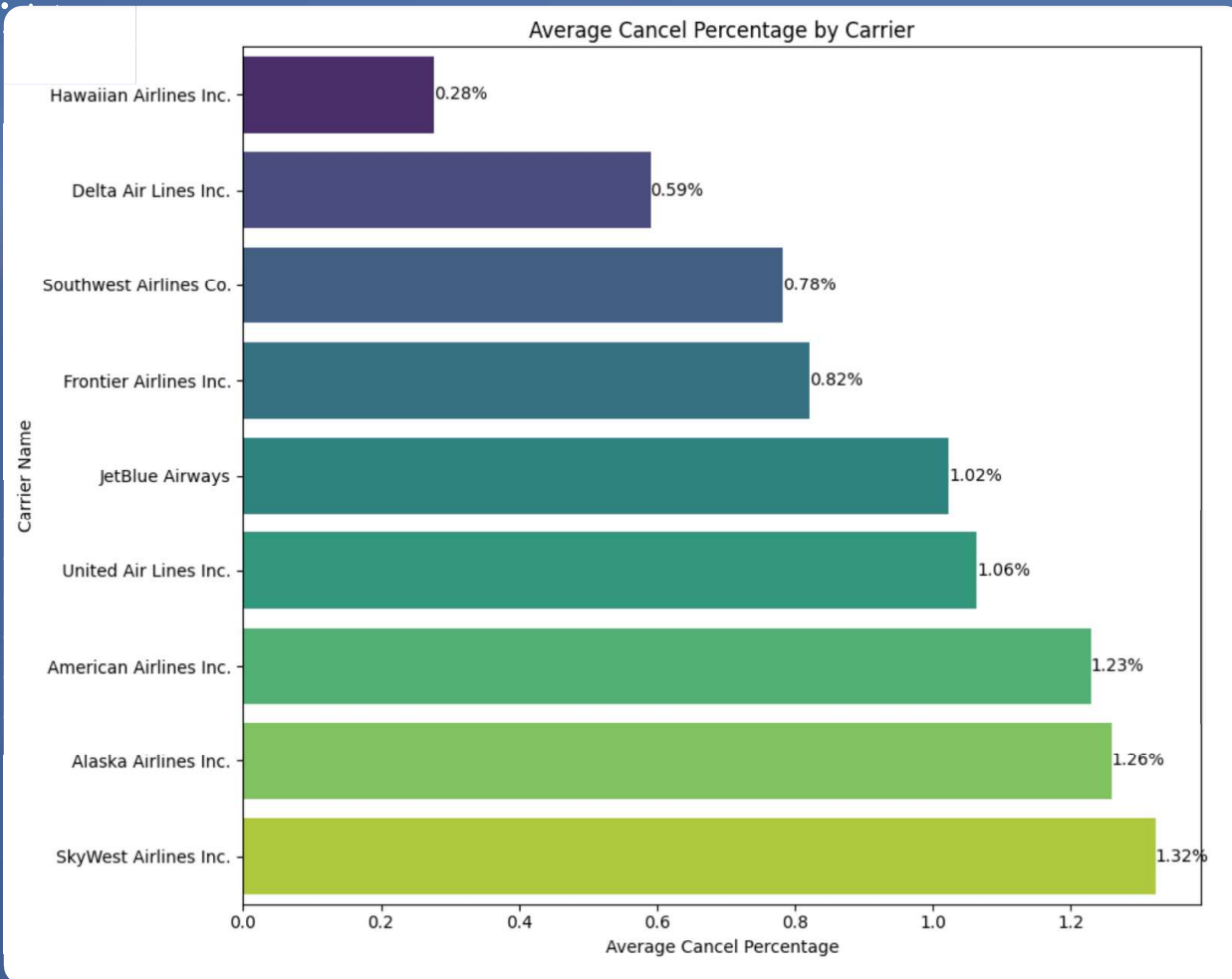
A “reportable” delay is one where the flight arrives 15 minutes or later after its scheduled arrival time.

Delay Count Over Time



DELAY ANALYSIS

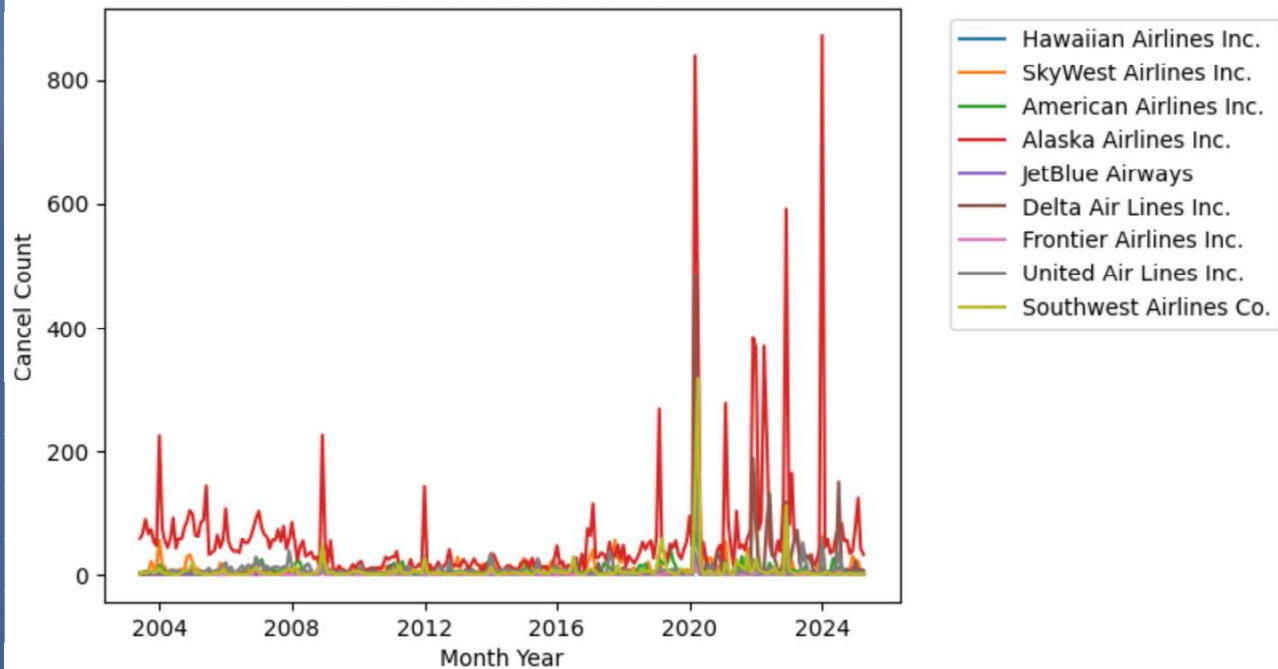
The sharp drop in airline delays during 2020 was largely due to COVID-19. With travel restrictions in place and far fewer flights operating, the system experienced minimal congestion, making delays less common.



CANCELLATION ANALYSIS

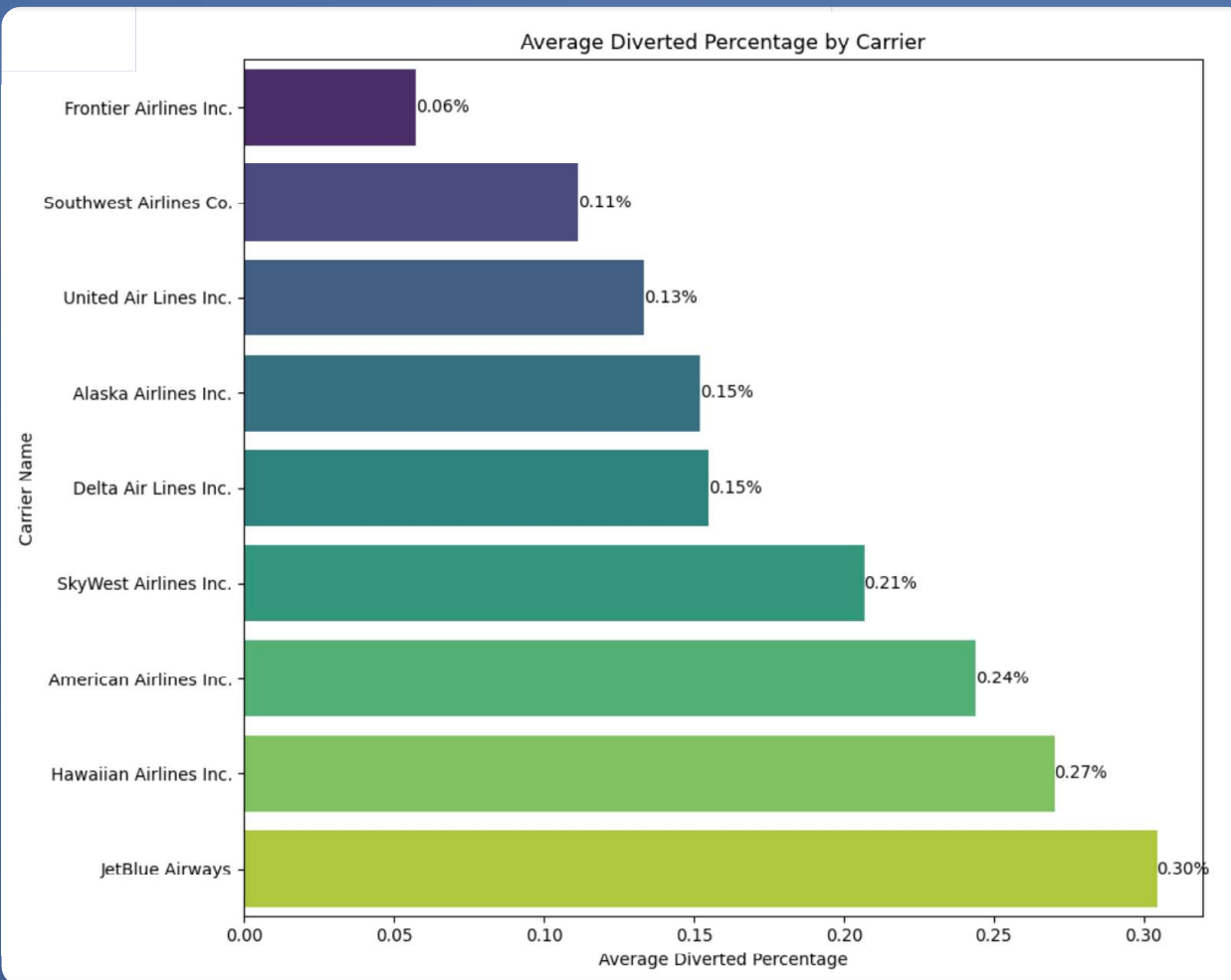
A scheduled flight is not operated at all, and the airline does not complete the planned transportation of passengers to the intended destination.

Cancel Count Over Time



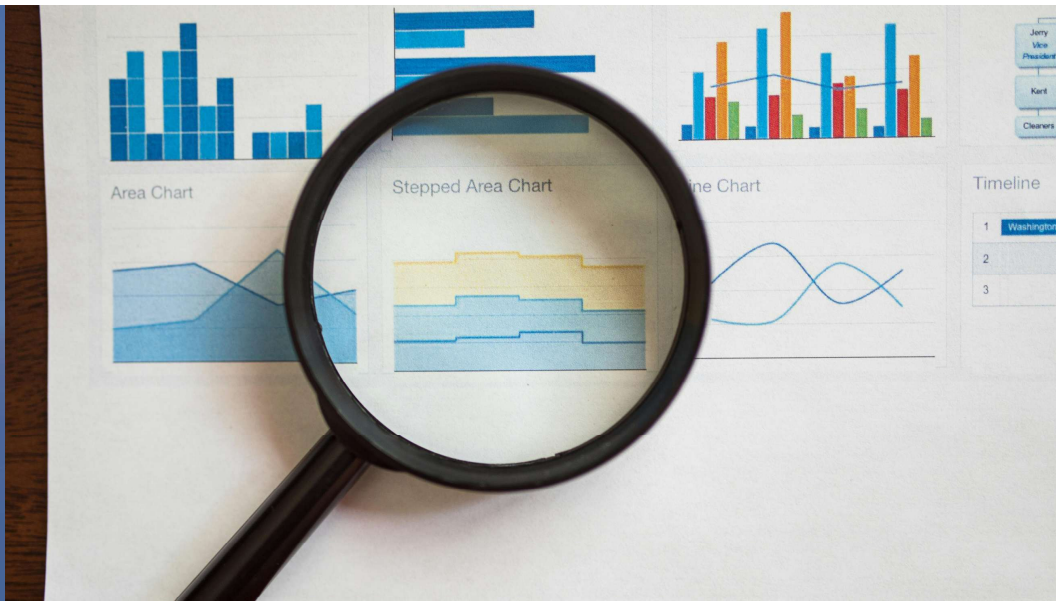
CANCELLATION ANALYSIS

- Staffing shortages – pilots, crew, and ground staff still below pre-COVID levels.
- Over-scheduling – airlines added too many flights for their available resources.
- Aircraft & parts delays – supply chain issues slowed repairs and deliveries.
- Air traffic control shortages – fewer controllers caused bottlenecks.
- More extreme weather – storms, heat waves, and smoke disrupted schedules.
- Demand surge – “revenge travel” overwhelmed an already fragile system.



DIVERSION ANALYSIS

Flights unable to reach their planned destination due to mechanical issues, in-flight emergencies, etc.



BUSINESS INSIGHTS

PRACTICAL BUSINESS IMPLICATION OF THIS ANALYSIS

- Airline performance benchmarking
- Customer experience and satisfaction
- Operational bottlenecks
- Cost impact analysis
- Fleet management and maintenance
- Seasonal and external factors

RSME

Delays:

Test RMSE: 9.9217

Train RMSE: 10.6261

Cancels:

Test RSME: 3.3578

Train RMSE: 2.3228

Diverted:

Test: 0.3701

Train RMSE: 0.6118

- More data on delays
- Less on cancels and diversions, making it more difficult to predict for this dataset
- RMSE numbers are low, but the difference between the test and training data for cancels and diversions is sizable



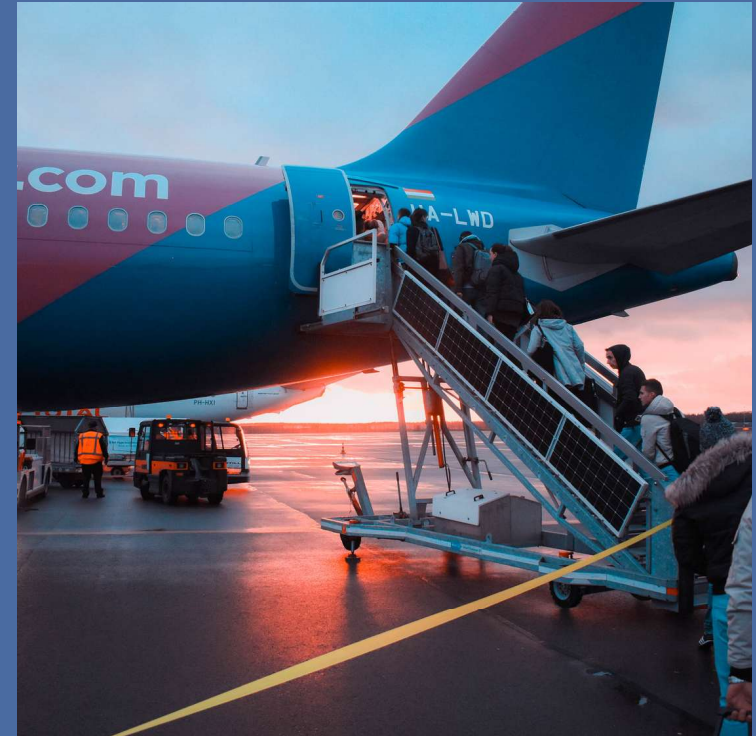
AI USE

- Gemini was resourceful given that it predicted the code based on the comment given in the code block.
- Saves time vs having to lookup the syntax on Stack Overflow.
- Gemini also provided code when prompted how to clean data such as consolidating the variation of carrier names.



TAKEAWAYS FOR FUTURE PROJECTS

- Consider other factors such as weather, National Air System (NAS), or security that could impact flight errors
- Forecasting data using seasonality parameters
- Improve data selection process





THANK YOU